

Local Skolemization

Predicate Logic with Anaphora, Calmed Down

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Abstract. Predicate Logic with Anaphora (PLA) (Dekker, 1994) is the compositional treatment of dynamic semantics whose particular appeal is firm logical foundations, conciseness and minimalism: conservatively extending the ordinary first-order logic just enough to describe discourse anaphora and donkey sentences, while maintaining the standard notions of binding.

In this paper we show that the first-order logic as is, *without any extensions*, is capable of clearly and concisely describing the same anaphoric phenomena: intra- and inter-sentential pronominal anaphora, bound-variable anaphora, conditional and relative-clause donkey sentences, quantificational subordination. Our approach is static, simply-typed, first-order and entirely orthodox. It cleanly separates syntax-semantics interface from the pragmatics of pronoun resolution.

The key idea is the familiar Skolemization, extended to implication formulas without requiring their normalization.

Predicate Logic with Anaphora (PLA) was proposed by Dekker [3] as a compositional and rigorous formulation of discourse representation theory with fully orthodox notions of scope and binding (unlike its predecessor Dynamic Predicate Logic (DPL) [4]) and as a proper, *dynamic*, extension of ordinary predicate logic. Its main idea is treating (nominal) pronouns as a separate category of terms, distinct from both constants and variables. In stark difference from DPL, pronouns are not subject to binding. Like DPL, PLA is a dynamic logic theory through and through.

In PLA, pronouns are represented by a distinct category of terms p_i where i refers to the i -th recently introduced antecedent. Therefore, to write the logical formula representing the meaning of a sentence, we have to know which pronoun refers to which antecedent, and in which order antecedents are introduced. We need to know, hence, not just static but dynamic context. In effect, we have to perform the pronoun resolution beforehand, that is, before the semantic analysis. The resolution must then rely solely on syntax, which is generally impossible. For example, should the meaning of “He beats it” be expressed by the formula $Beat(p_0, p_1)$ or $Beat(p_1, p_0)$, or perhaps $Beat(p_0, p_2)$? We cannot tell without the context: we cannot assign the meaning to a sentence or its constituent in isolation, that is, compositionally.

Dekker [3] has identified the principal stumbling block for any logical theory of discourse anaphora: if an antecedent for a pronoun is an indefinite in an earlier sentence in the discourse and the sentence is represented by a closed existential formula, there is no way to refer to that antecedent. We would need an “existential disclosure” so to speak [2]. Dynamic semantics is one way of achieving such disclosure; Σ -types in Dependent Type Semantics is another.

In this paper we observe that the “existential disclosure” can be achieved by Skolemization, which is the tried-and-true way to replace an existential with its witness. It was proposed back in 1920s and is widely used in logic meta-theory (e.g., in Henkin method for proving completeness) as well as in automated theorem provers.

Based on this observation, we present a theory of intra- and inter-sentential pronominal anaphora, bound-variable anaphora, conditional and relative-clause donkey sen-

tences and quantificational subordination that is based *solely* on the orthodox predicate logic, with no extensions and with the standard model-theoretic semantics. It is static, with the standard notions of scope and binding.

Because our approach is fully classical, we can use the existing tools such as automated theorem provers (E-prover, etc.) or Prolog.

Our goal is to explain the mechanism of anaphoric references in the simplest and rigorous logical terms; give the orthodox classical-logic account of discourse anaphora and donkey sentences; reproduce the empirical data (e.g., the absence of reference in negated, implication and universally quantified sentences). One may use our approach with any semantic framework of one's choosing: PTQ, QR, ACG, continuation/monad/applicative semantics, etc. Pronoun resolution is not baked-in but becomes an explicit step, after the semantic analysis.

For the lack of space, and just like PLA, we restrict our attention in this paper to singular pronouns whose referent is a bound variable. We use the completely standard classical first-order logic (FOL). To save space, we elide formal definitions: see [6] for reference and notation.

1 Discourse and Bound-Variable Anaphora

In our approach, truth conditions are represented as a predicate logic formula, derived using whatever semantic theory (syntax-semantic interface) one prefers. *A pronoun is represented by the term $F(x_1, \dots, x_n)$ where F is a fresh function symbol (constant) and $x_1, \dots, x_n$ are the variables in scope.* The following examples illustrate. The simplest is

- (1) A man walked. John saw him.
- (2) $(\exists x. \text{Man}(x) \wedge \text{Walked}(x)) \wedge \text{Saw}(F_1, \text{John})$

The meaning of the two-sentence discourse (1) is straightforwardly expressed by (2). Since the second sentence has no quantifiers or free variables, ‘him’ is represented by a constant: F_1 .

Recall, an *interpretation* of a logical formula ϕ like (2) is specified by a non-empty set D (domain of discourse) and an interpretation function I that maps every constant to a member of D , every arity- n function symbol to a map $D^n \mapsto D$ and every arity- n predicate letter to a n -place relation over D . *Satisfaction* is a relation between ϕ , (D, I) , and a variable assignment V mapping variables to elements of D – notated as $(D, I) \models_V \phi$ and read as (D, I) satisfies ϕ (under the assignment V).

Formula (2) has a variety of satisfying interpretations. In some of them x is interpreted as the same domain element as F_1 , in some it is not. That corresponds to the range of meanings of (1): the pronoun may be resolved to the object introduced by the indefinite, or it may remain unresolved (that is, refer to something else in the discourse or common knowledge, or just left to the speaker imagination.)

Skolemization is replacing existentials by constants (generally, function symbols): the witnesses [7]. The Skolemized version of (2) is

- (3) $(\text{Man}(C_1) \wedge \text{Walked}(C_1)) \wedge \text{Saw}(F_1, \text{John})$

where C_1 is a fresh constant. A Skolemized formula is equi-satisfiable with the original one: if there is interpretation of the original, there is an interpretation of the Skolemized formula, and vice versa.

Formula (3) has the same variety of interpretations. In some, C_1 and F_1 are mapped by I to the same element of D , in some they are not. If we are interested only in the former, i.e., resolved, interpretations, we can state so as the side-condition (narrowing of the interpretations): “ I where $I(C_1) = I(F_1)$ ”. Alternatively, we may adjoin to (3) an equality

$$(4) \quad (Man(C_1) \wedge Walked(C_1)) \wedge Saw(F_1, John) \wedge C_1 = F_1$$

Now, only interpretations with ‘him’ referring to the walking man satisfy (4). The restricted formula is, of course, not equivalent to (3); rather, the former implies the latter. The constraint $C_1 = F_1$ is quite like the co-indexing notation commonly used in semantic analyses.

How exactly to resolve a pronoun to an available antecedent – that is, which Skolem constant to pick out of several available – is in the domain of pragmatics and out of scope for us. It is in scope for Butler [1], who describes the procedure in more detail.

The next example: the original sentence, the corresponding logical formula, its Skolemization, and the augmented formula reflecting the pronoun resolution.

(5) A man walked. A passer-by saw him.

$$(6) \quad (\exists x. Man(x) \wedge Walked(x)) \wedge (\exists y. PB(y) \wedge Saw(F_1(y), y))$$

$$(7) \quad (Man(C_1) \wedge Walked(C_1)) \wedge (PB(C_2) \wedge Saw(F_1(C_2), C_2))$$

$$(8) \quad (Man(C_1) \wedge Walked(C_1)) \wedge (PB(C_2) \wedge Saw(F_1(C_2), C_2)) \wedge \forall x. F_1(x) = C_1$$

We did not have to introduce the arity-1 function symbol F_1 : a constant would have sufficed. F_1 is interpreted as a function that maps an element of D , such as $I(C_2)$ to another element of D , which may be equal to $I(C_1)$, or not. To make sure it is equal we may adjoin the formula as in (8), which hence expresses the resolution of the pronoun. In the related “A colleague left. He said he was ill.” the two pronouns are represented by two different constants F_1 and F_2 , which may however be interpreted the same, and specified by the resolution condition $F_1 = F_2$. When deriving the logical formula we, unlike PLA, do not have to decide if the two ‘he’ refer to the same person or not. We defer the decision to the pragmatic resolution step.

The next example, described similarly:

A man walked. Every passer-by saw him.

$$(\exists x. Man(x) \wedge Walked(x)) \wedge (\forall y. PB(y) \rightarrow Saw(F_1(y), y))$$

$$(Man(C_1) \wedge Walked(C_1)) \wedge (\forall y. PB(y) \rightarrow Saw(F_1(y), y))$$

$$(Man(C_1) \wedge Walked(C_1)) \wedge (\forall y. PB(y) \rightarrow Saw(F_1(y), y)) \wedge \forall x. F_1(x) = C_1$$

Only existentials are subject to Skolemization; the universal quantifier is left as is [7]. A slight variation of the example

Every man walked. He whistled.

$$(\forall x. Man(x) \rightarrow Walked(x)) \wedge Whistled(F_1)$$

has only the universal quantification, to which Skolemization does not apply. There is nothing therefore to identify F_1 with.

- (9) No man walked. He whistled.
 (10) $(\neg\exists x. \text{Man}(x) \wedge \text{Walked}(x)) \wedge \text{Whistled}(F_1)$
 (11) $= (\forall x. \neg(\text{Man}(x) \wedge \text{Walked}(x))) \wedge \text{Whistled}(F_1)$

Skolemization applies only to formulas in the prenex form: all quantifiers are at the front. $\neg\exists x. \text{Man}(x) \wedge \text{Walked}(x)$ is not in the prenex form – but it can be converted to one: see (11). The result has no existentials and hence no Skolemization is possible. The pronoun remains unresolved. Another way to understand negation is as an implication to false. The implication is dealt in §2.

The example borrowed from [5]

Exactly one man loves Annie. He is rich.

$$\text{Man}(C_1) \wedge \text{Loves}(\text{Annie}, C_1) \wedge (\forall x. x \neq C_1 \wedge \neg(\text{Man}(x) \wedge \text{Loves}(\text{Annie}, x))) \\ \wedge \text{Rich}(F_1) \wedge F_1 = C_1$$

shows that our approach does not over-generate: it does *not* give a reading that there are possibly several men who love Annie, but only one of them is rich.

Intra-sentential bound-variable anaphora is straightforward:

Every male lawyer believes he is smart.

$$\forall x. \text{MaleLawyer}(x) \rightarrow \text{Believe}(\text{Smart}(F_1(x)), x)$$

$$\forall x. \text{MaleLawyer}(x) \rightarrow \text{Believe}(\text{Smart}(F_1(x)), x) \wedge \forall x. F_1(x) = x$$

That (13) is not a felicitous continuation of (12) but (14) is an example of quantificational subordination [5].

- (12) Bryan bought a bottle of wine at every store in the neighborhood.
 (13) #It was a pinot noir.
 (14) It was always a pinot noir.

The Skolem function for “a bottle of wine” must be an arity-1 function symbol, since it is in scope of “every store”. Whereas in (13) ‘it’ is represented by a constant symbol, in (14) it is also an arity-1 function symbol, being in the scope of the universal ‘always’. The two function symbols may then be identified.

2 Donkey Sentences and Local Skolemization

Donkey sentences are one of the main motivations for developing discourse representation theory and dynamic semantics. The discourse-anaphora donkey sentence [3, (1)] is straightforward, as expected.

- (15) A farmer owns a donkey. He beats it.
 (16) $\text{Farmer}(C_1) \wedge \text{Donkey}(C_2) \wedge \text{Owns}(C_2, C_1) \wedge \text{Beat}(F_2, F_1)$

(We could also adjoin $Man(F_1) \wedge NonHuman(F_2)$.) The resolution easily identifies F_i with C_i .

The conditional donkey sentence is far more interesting:

(17) If a farmer owns a donkey he beats it.

(18) $(\exists f. \exists d. Farmer(f) \wedge Donkey(d) \wedge Own(d, f)) \rightarrow Beat(F_2, F_1)$

Since the formula is not in prenex form, the Skolemization does not apply. We have to look carefully at implication.

Recall the correctness of Skolemization: a Skolemized formula is equi-satisfiable with the original one. For example,

(19) $(D, I) \models \exists x. \phi(x) \text{ iff } \exists d \in D. (D, I \cup \{C \mapsto d\}) \models \phi(C)$

where C is fresh (does not occur in ϕ). For implication, we have

(20)
$$\begin{aligned} (D, I) &\models (\exists x. \phi(x)) \rightarrow \psi \\ &\equiv (D, I) \models \exists x. \phi(x) \text{ implies } (D, I) \models \psi \\ &\equiv \forall d \in D. ((D, I \cup \{C \mapsto d\}) \models \phi(C) \text{ implies } (D, I) \models \psi) \\ &\equiv \forall d \in D. (D, I \cup \{C \mapsto d\}) \models \phi(C) \rightarrow \psi \end{aligned}$$

We took advantage of the fact that C is fresh: occurs neither in ϕ nor ψ . Therefore, if $(D, I) \models \psi$, extending I with the interpretation of C preserves the satisfaction.

The implication $\phi(C) \rightarrow \psi$ in (20) may be true because ψ is satisfied independently of whether there is $d \in D$ that has the property described by ϕ . On the other hand, the satisfaction of ψ may depend on the existence of such d : that is, there is some constant in ψ that is interpreted as d . Let F be such constant. The sufficient condition for (20) may then be written as

(21)
$$\begin{aligned} (20) &\Leftarrow \forall d \in D. (D, I \cup \{F \mapsto d, C \mapsto d\}) \models \phi(C) \rightarrow \psi \\ &\equiv \forall d \in D. (D, I \cup \{F \mapsto d, C \mapsto d\}) \models \phi(C) \rightarrow \psi \wedge F = C \end{aligned}$$

Since the notation became heavy, we abstract away D and I in (20) and introduce *local Skolemization* $\mathcal{AC}. \phi(C) \rightarrow \psi$, to be read as “for all domain elements that C represents, if $\phi(C)$ is satisfied then so is ψ . The formula ψ may refer to that domain element – in effect, refer to C . We must stress that $\mathcal{AC}. -$ is not a new quantifier: the new notation is a mere shorthand for (20). What universally quantified is a domain element; C is a proxy for it. In effect, however, $\mathcal{AC}. -$ has the universal force. Furthermore, since d that is associated with C in (20) is universally quantified, it cannot be referred to outside the local Skolemization scope.

We may revisit (11) and write it as

(22) No man walked. He whistled.

(23) $((\exists x. Man(x) \wedge Walked(x)) \rightarrow \perp) \wedge Whistled(F_1)$

(24) $= (\mathcal{AC}. (Man(C) \wedge Walked(C)) \rightarrow \perp) \wedge Whistled(F_1)$

Since C is a local Skolem constant, F_1 cannot be identified with it.

In the local Skolemization notation, the conditional donkey sentence (18) can be represented by (26) and resolved as (27).

(25) If a farmer owns a donkey he beats it.

(26) $\mathcal{AC}_F. \mathcal{AC}_D. \text{Farmer}(C_F) \wedge \text{Donkey}(C_D) \wedge \text{Own}(C_D, C_F) \rightarrow \text{Beat}(F_2, F_1)$

$\Leftarrow \mathcal{AC}_F. \mathcal{AC}_D. \text{Farmer}(C_F) \wedge \text{Donkey}(C_D) \wedge \text{Own}(C_D, C_F)$

(27) $\rightarrow \text{Beat}(F_2, F_1) \wedge F_1 = C_F \wedge F_2 = C_D$

For the relative-clause donkey sentence we have

(28) Every farmer who owns a donkey beats it.

(29) $\forall f. \text{Farmer}(f) \rightarrow (\exists d. \text{Donkey}(d) \wedge \text{Owns}(d, f)) \rightarrow \text{Beat}(F_1(f), f)$

(30) $\forall f. \text{Farmer}(f) \rightarrow \mathcal{AC}_1. \text{Donkey}(C_1(f)) \wedge \text{Owns}(C_1(f), f) \rightarrow \text{Beat}(F_1(f), f)$

The function symbol F_1 may be identified with C_1 , which gives the (desired) strong reading of the sentence.

For “Some farmer who owns a donkey beats it.” we get the expected weak (existential) reading.

3 Related work

PLA and its exposition of the stumbling block has been an inspiration for this work, stimulating the search for the minimal workaround. We also wanted to avoid the baking-in of the pronoun resolution, which is responsible for many PLA problems: inability to write the denotation of a sentence in isolation, complicated weakening and transitivity laws and “some additional bookkeeping”. Like in PLA, but clearly unlike DPL, pronouns are represented by constants – not variables. Unlike PLA, our constants are ordinary, rather than a special category with a special, dynamic, treatment within a model.

Any dynamic semantic theory presupposes an “arrow of time” and hence the direction of anaphora resolution. Such theory hence has problems with inverse anaphora, which, albeit rare, does occur. Our semantic analysis is static: it represents pronouns but defers their resolution to pragmatics.

Like us, Butler (in the long line of research, see [1] for the latest) treats antecedent and pronouns as entities, to be identified in the resolution procedure. Unlike us, the resolution procedure is one of the main topic of Butler’s research and described in great detail. Butler also employs Skolemization as an intermediary step.

D-type theories of conditional donkey sentences often employ situation semantics: assume an implicit “always” quantifier, responsible for the universal force. On our analysis, the universal force is not a postulate: it comes out as a pure logical consequence. We analyze ‘if’ as a ordinary logical implication.

In conclusion, We have described how the ordinary classical predicate logic by itself, without any extensions, can represent a variety of discourse and donkey anaphora. We employ the standard Skolemization, and extend it to apply to implication without prior normalization (which makes it applicable, incidentally, to intuitionistic logic).

For future work, we would like to analyze further anaphoric phenomena: paycheck pronouns, epistemic modalities, telescoping and parascoping.

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